Two Distributions, Opposite Verdicts: Feature Provenance in Cross-Dataset Federated Intrusion Detection

Hoang Tu Trinh and Tien Thanh Cao
Faculty of Information Technology
Ho Chi Minh City University of Foreign Languages – Information Technology
Ho Chi Minh City, Vietnam
Email: tht.csec2005@gmail.com, thanhct@huflit.edu.vn

***Abstract*—Cross-dataset intrusion-detection results are usually attributed to distribution shift between corpora. We show that for a fixed source model and a fixed set of target flows, the published verdict can instead be decided by which public CSV distribution of the target dataset supplied the features. We train logistic-regression FedAvg on InSDN in a five-client, five-seed setting and evaluate it zero-shot on CICIDS2017. The primary experiment is a row-matched counterfactual over a fixed 207,463-row target split, scored on paired stratified 30,000-row samples: identical rows in identical order, with identical labels, identical attack prevalence, and seven bit-identical features, in which only `Protocol` and `Source Port` carry the surrogates of one `MachineLearningCVE`-compatible harmonization. ROC-AUC falls from 0.7357 to 0.3545 and MCC from $+0.2511$ to -0.1915, a below-random cross-domain ranking that could otherwise be misread as model-transfer failure. The actual `MachineLearningCVE` release shows the same direction and similar magnitude (0.3441, MCC -0.2698), and a centralized model on the same records behaves the same way ($0.7490 \rightarrow 0.3544$), so the effect is not an artifact of FedAvg aggregation. Masking localizes the ranking reversal to protocol, whose learned contribution changes sign (-0.1438 versus $+0.2647$ ROC-AUC) while its marginal Kolmogorov–Smirnov distance is the *smaller* of the two; on identical rows, dropping the unavailable field is markedly safer than reconstructing it (0.5157 versus 0.3545). A label-free preflight check flags the substituted port on every seed, but none of the tested marginal distance rules separates the reconstructed protocol, so feature provenance has to be reported per field.**

***Index Terms*—Federated learning, intrusion detection, cross-dataset evaluation, feature provenance, reproducibility.**

I. Introduction

Federated learning (FL) lets participating sites improve a shared intrusion detector without centralizing raw records [1], [2]. Cross-dataset tests are the standard check on whether such a model generalizes beyond its training corpus, and they routinely report severe degradation [3]–[6], normally explained by differences in traffic, attack mix, or labeling.

This paper reports a different cause. Widely used corpora ship in more than one flow-feature distribution, and those distributions do not expose the same columns. CICIDS2017 has a `TrafficLabelling` release retaining `Protocol` and `Source Port`, and a `MachineLearningCVE` release omitting both, so anyone harmonizing feature names against the second has to decide how to handle two absent fields. We show that this choice, not the target traffic, can decide whether transfer looks usable or below random.

We ask three questions: (RQ1) does FedAvg improve over matched centralized and local controls; (RQ2) how does an explicit non-IID partition affect ranking and fixed-threshold measures; and (RQ3) does the verdict depend on which public distribution supplied the target features, and which field drives that dependence? FedAvg is the evaluation setting here, not the mechanism we claim.

Our contributions are: (i) matched centralized, local-only, IID, and non-IID controls with paired seed-level reporting; (ii) a row-matched counterfactual that isolates the mechanism, holding the target rows, their order, their labels and all seven remaining features bit-identical while only the two provenance-sensitive fields change, paired with the real `MachineLearningCVE` release as an ecological check and with a centralized control showing the effect is not specific to federated averaging; and (iii) a masking diagnostic that localizes the ranking reversal to protocol, a quantified comparison showing that dropping an unavailable field is safer than reconstructing it, and a label-free preflight screen that catches placeholder substitution but not this reconstruction, which is why per-field provenance must be reported.

II. Related Work and Research Gap

A. Federated intrusion detection and client heterogeneity

FedAvg established sample-weighted aggregation of local updates [1], and security surveys distinguish distributed optimization from formal privacy or attack robustness [2]. Because non-IID clients cause objective drift, FedProx adds a proximal term [7], SCAFFOLD corrects drift with control variates [8], and FedNova normalizes unequal local work [9]. FL-IDS work spans linear scanning detectors [10], 5G experiments with centralized and local controls [11], and deep or distillation-based defenses [12], [13]. Dataset-centric FL evaluation reports in-domain scores approaching centralized ones while cross-dataset transfer degrades sharply even after label and feature spaces are aligned [14]. We keep plain FedAvg and logistic regression; the aggregation variants above are a next step, not a claim here.

B. Cross-dataset NIDS and feature semantics

Dataset surveys show that collection environment, labeling, traffic generation, and flow extraction constrain valid comparisons [15]. Inter-dataset studies report large degradation even between related CIC corpora [3]; XeNIDS formalizes cross-evaluation and warns against assuming interchangeable flow fields [4]. Later work finds asymmetric transfer across four datasets [5] and near-perfect within-domain yet poor cross-dataset performance [6]. Standardized NetFlow feature sets improve generalization but feature influence stays dataset-dependent [17]. Closest to our setting, Demirbaş Paray and Aydos harmonize CICIDS2017 and InSDN/OVS into a common schema for multi-domain federated IDS, on the ground that both were produced by the same extractor, so fields such as flow duration and packet length "carry identical semantic and technical definitions" [16]. We do not dispute the consistency of those shared fields. Our result shows that shared extractor lineage is nonetheless insufficient to validate *every* field of a harmonized schema, because release-specific columns of the same corpus may be absent and reconstructed.

C. Dataset defects, tooling, and release variants

CICIDS2017 has documented defects rather than merely different statistics. Engelen et al. report flow-construction and labeling faults [18]; Lanvin et al. find packet misordering, duplication, and an unlabeled port scan, and measure the resulting change in detection performance [19]; Rosay et al. audit the released flows and report duplicated features, miscalculated features, and *wrong protocol detection*, noting that the published CSVs cannot be regenerated from the PCAPs with any available extractor version [20].

This literature establishes that a corpus can be internally faulty and that two corpora are rarely comparable. Recent heterogeneous-domain benchmarking goes further and makes the feature interface itself explicit, through intersection and union representation contracts that deterministically materialize fields absent from a domain [21]; it aligns *distinct* datasets. Our gap is narrower and, to our knowledge, unquantified: whether provenance differences between public releases of the *same* corpus can invert the sign of a cross-dataset conclusion while rows, labels, prevalence and every unaffected field are held fixed. Two releases cannot settle this directly, because their row sets and prevalence also differ. We therefore construct a counterfactual on one fixed row set so that feature provenance is the only variable, then report the mechanism and ask what an automated check could have caught.

III. Method

FedAvg aggregates locally updated parameter vectors as $\mathbf{w}_{t+1} = \sum_{k=1}^{K}(n_k/\sum_j n_j)\,\mathbf{w}_{t+1}^{k}$ with $K = 5$ clients and local sample counts n_k [1]. Each client runs one local pass of binary logistic regression per round, giving $p(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$. The model is a transparent tabular baseline chosen so that the cross-view comparison cannot be attributed to architecture.

Server and clients begin from the same zero coefficient/intercept state, so no client record initializes the model, and a "local epoch" is one full `partial_fit` pass, not a mini-batch count. Centralized LR uses the same optimizer and 20 passes; local-only LR keeps independent client states and averages their test metrics. The source-training mean and standard deviation define each z-score and are retained unchanged for the InSDN test partition and all three target views; non-finite rates and missing values become zero before that fit.

A. Feature-shift diagnostic

For source feature j with empirical CDF F_{sj} and target CDF F_{tj}, we record the two-sample Kolmogorov–Smirnov distance $D_j = \sup_x |F_{sj}(x) - F_{tj}(x)|$. This marginal statistic cannot reveal whether a learned association changes sign. We therefore keep the trained IID global model fixed, replace one standardized feature by zero (the source-training mean) in both test domains, and compute $\Delta A_j = A_t^{\mathrm{mask}(j)} - A_t^{\mathrm{full}}$ for target ROC-AUC. A positive ΔA_j identifies a feature whose learned contribution harms target ranking; a negative one identifies a feature carrying transferable signal. For a linear score the masked logit is exactly $\mathbf{w}^\top\mathbf{x} - w_j x_j$, so the diagnostic is cheap and exact here, and it applies unchanged to nonlinear scores. It does, however, consume target labels, so it audits a finished experiment and cannot protect one in progress.

B. Label-free provenance screen

We therefore ask what can be established before any target label exists, from the source-training column x_j and the unlabelled target column $\tilde{x}_j$ alone. The check flags a column when either

$$\frac{\mathrm{sd}(\tilde{x}_j)}{\mathrm{sd}(x_j)} < \tau_v \quad \text{or} \quad \Pr[\tilde{x}_j \notin [\min x_j, \max x_j]] > \tau_r, \qquad (1)$$

with $\tau_v = \tau_r = 0.01$. The first condition detects a column that has lost essentially all variation, the second a column whose values were never seen in training; a placeholder written for an absent field triggers both. The thresholds are deliberately one-sided, since a *large* dispersion ratio is ordinary shift rather than missing data. For integer-coded fields we also record two candidate rules: the total-variation distance between category proportions, and whether any category holding over 5% of source mass falls below 1% of target mass. The label-aware ΔA_j is the reference signal these label-free rules are compared against, not causal ground truth: with correlated features, masking reports how the model relies on a column, not how much it was corrupted.

IV. Experimental Protocol

A. Source data and the three target views

We use InSDN [22] for training and within-domain testing. The prepared InSDN file contains 343,889 records (68,424 `normal`; 275,465 `anomaly`) and carries true IANA protocol

TABLE I
THE THREE TARGET VIEWS OF THE SAME CAPTURE FILES. CF IS ROW-MATCHED TO TL BY CONSTRUCTION; ML IS THE REAL RELEASE AND DIFFERS IN ROW SET AS WELL.

	TL (original)	CF (counterfactual)	ML (MLCVE)
Protocol field	original	flag surrogate	flag surrogate
Protocol values	$\{0, 6, 17\}$	$\{0, 6\}$	$\{0, 6\}$
Source port	original	sentinel -1	sentinel -1
Cleaned flows	1,037,315	1,037,315	880,176
Fixed test split	207,463	207,463	176,036
Sampled benign/attack	21,705 / 8,295	21,705 / 8,295	22,577 / 7,423
Attack prevalence	0.2765	0.2765	0.2474
Row-matched to TL	—	yes	no

numbers $\{0, 6, 17\}$. Each seed draws a stratified 30,000-record sample, split into 24,000 train and 6,000 test records, and draws 30,000 records from each target view. Because TL and CF share their row order and labels, one seeded draw selects the same rows from both, so every TL–CF comparison below is paired row by row; the evaluation asserts this before scoring.

All three target views describe the *same* CICIDS2017 capture files [23]: Monday benign, Friday DDoS, Friday PortScan. The TL view is rebuilt from `TrafficLabelling` (`GeneratedLabelledFlows`) and keeps the original `Protocol` and `Source Port` fields, whereas ML uses `MachineLearningCVE` (`MachineLearningCSV`); the artifact records each field's source and read-or-reconstructed status. Neither column exists in `MachineLearningCVE`, so preserving the nine-field schema requires handling both absent fields; the release forces the missingness, not any particular remedy. We evaluate one plausible but lossy rule, chosen as a controlled failure mechanism rather than as documented practice: protocol becomes TCP (6) when any TCP flag is positive and 0 otherwise, and every source port becomes -1. We do not claim anyone harmonizes this way; it is a choice, not a standard, and Section V-B contrasts it with dropping the field altogether.

Comparing TL against ML alone would be confounded, because the two releases survive cleaning and de-duplication differently and end up with different row sets and attack prevalence (Table I). The CF view removes that confound: it is derived from the TL test split itself, the same 207,463 rows in the same order with the same labels and same seven remaining features, and only `ip_proto` and `tp_src` overwritten by the ML surrogates. Deriving it from the finished row set rather than re-running the cleaner is essential, since the missing-value filter and the de-duplication both key on the two columns being changed and would otherwise select different rows. The preparation script asserts that the unaffected columns and labels are bit-identical and that exactly two columns differ; the evaluation script repeats that assertion after seeded subsampling, and either failure aborts the run. The surrogate assigns a different protocol to 30.32% of rows, since UDP (17) has no representation in the flag-derived rule.

The complete input is nine columns, identically named in the prepared InSDN CSV and in every target view and referenced by name rather than by position: `ip_proto`, `tp_src`, `tp_dst`, `packet_count`, `byte_count`, `duration_sec` (seconds), `packet_count_per_sec`, `byte_count_per_sec`, and `packet_size_avg` (bytes). All nine are z-scored; the two rate columns have non-finite values mapped to zero beforehand; ports span 0–65,535 except for the sentinel -1; protocol is integer-coded, and is one-hot encoded only in the corresponding ablation, which leaves every other column in place, while a separate ablation removes both ports. The prepared CSVs also carry a `flow_duration` column identical to `duration_sec` in every row, an instance of the feature duplication reported for these releases [20]; it is excluded so no coefficient is split across duplicated inputs. Labels are binary: InSDN `normal`$\rightarrow$ 0 and `anomaly`$\rightarrow$ 1, giving mean sampled counts of 5,969 benign and 24,031 attack, while in every target view `normal`$\rightarrow$ 0 and `ddos`/`portscan`$\rightarrow$ 1. Sampling follows label conversion and numeric cleanup and is stratified to retain prevalence. No source string, run identifier or label field reaches the classifier.

`StandardScaler` is fit once on the InSDN training features and applied unchanged to the InSDN test partition and all three target views; no target-domain fit, fine-tuning, or threshold selection is performed.

B. *Partitions and measures*

For IID FL, shuffled training records are divided equally among five clients; for non-IID FL, each class is allocated by a Dirichlet distribution with $\alpha = 0.5$. Both run 20 rounds fixed *a priori*, and no early stopping, model selection or tuning uses the test set. The random PR baseline equals the attack prevalence: 0.801 in InSDN and the view values in Table I.

We compute precision, recall, F1, ROC-AUC, PR-AUC, balanced accuracy, MCC and confusion matrices; space permits five of them in the tables, and the artifact CSVs carry all of them per seed. Predicted labels use the fixed 0.5 threshold; ROC-AUC and PR-AUC use continuous scores. Balanced accuracy stops the majority attack class from dominating, and the sign of MCC makes a below-random operating point visible.

IID partitioning gives every client 4,800 records and retains class proportions, whereas the Dirichlet split produces severe skew: at seed 11, client sizes range from 1,669 to 11,950 records and the benign share from 0.5% to 38%. Centralized LR uses the identical 24,000 training records; local-only LR trains five independent client models on the IID partition and reports their unweighted metric mean, a descriptive reference over independently deployed clients, not an ensemble or a global predictor.

The classifier is a scikit-learn `SGDClassifier` with `log_loss`, constant-rate SGD at $\eta = 0.01$ and L2 $\alpha = 10^{-4}$, no class weighting, and a fixed 0.5 threshold; there are $K = 5$ clients, 20 rounds, one full local pass per round, seeds 11, 23, 42, 77 and 101, and infinities are mapped to NaN and then to zero before scaling. The pinned environment is Python 3.11 with NumPy 2.2.6, pandas 2.3.1 and scikit-learn 1.7.2, and `summary.json` records every one of these values alongside the results.

TABLE II

SOURCE-DOMAIN (INSDN) FIVE-SEED RESULTS; MEAN±STD.

Setting	F1	ROC-AUC	Bal. acc.	MCC
Central	0.9567±0.0033	0.9320±0.0038	0.8285±0.0149	0.7624±0.0199
Local-only	0.9588±0.0027	0.9288±0.0030	0.8367±0.0120	0.7751±0.0162
FedAvg IID	0.9580±0.0023	0.9324±0.0039	0.8355±0.0091	0.7709±0.0140
FedAvg non-IID	0.9556±0.0038	0.9336±0.0064	0.8136±0.0166	0.7562±0.0228

TABLE III

ZERO-SHOT CROSS-DATASET RESULTS, FIVE SEEDS, NO REFITTING. CF DIFFERS FROM TL IN EXACTLY TWO COLUMNS ON IDENTICAL ROWS.

Target view	F1	ROC-AUC	PR-AUC	Bal. acc.	MCC
FedAvg IID global model					
TL (original)	0.5012±0.0037	0.7357±0.0070	0.4520±0.0154	0.6403±0.0037	0.2511±0.0065
CF (row-matched)	0.3558±0.0026	0.3545±0.0031	0.2102±0.0009	0.4191±0.0037	−0.1915±0.0097
ML (real release)	0.2914±0.0018	0.3441±0.0040	0.1848±0.0010	0.3766±0.0023	−0.2698±0.0067
Centralized model, same records					
TL (original)	0.4950±0.0041	0.7490±0.0165	0.4656±0.0225	0.6336±0.0047	0.2392±0.0083
CF (row-matched)	0.3545±0.0020	0.3544±0.0033	0.2100±0.0010	0.4163±0.0045	−0.2004±0.0147
ML (real release)	0.2903±0.0041	0.3447±0.0048	0.1848±0.0012	0.3738±0.0074	−0.2782±0.0198

V. RESULTS

A. Source-domain controls

Table II reports mean±std over five seeded repeated holdouts. All four configurations reach F1 near 0.956–0.959, with FedAvg IID between the centralized and local-only controls and a gap to centralized smaller than the seed-to-seed deviations. Local-only has the highest mean F1, which is not evidence of superiority since it is an unweighted mean over five models on one test set. Non-IID allocation preserves ROC-AUC while reducing balanced accuracy and MCC, an operating-point effect under client class skew rather than uniformly worse ranking. Round 20 was fixed before evaluation; the saved history shows both partitions flat well before it. Across the four configurations attack recall is 0.9930–0.9995 while benign recall is only 0.6278–0.6793, which is why F1 and PR-AUC alone are insufficient and why Table II includes MCC; balanced accuracy is their mean. The non-IID model has the highest attack recall and the lowest benign recall, the operating-point shift behind its lower balanced accuracy. Per-seed values are in `five_seed_metrics.csv`: IID FedAvg F1 ranges from 0.9557 to 0.9606 and the non-IID range is wider (0.9510–0.9611).

B. Two fields reverse the verdict on identical rows

The seven-field intersection of Section V-C is our deployment-compatible cross-dataset configuration; the nine-field comparison here is a controlled mechanistic experiment isolating the cost of reconstructing unavailable fields, not a recommended interface. The same five IID FedAvg models are scored on all three views without refitting (Table III). On TL, transfer is clearly degraded relative to the source domain ($0.9324 \rightarrow 0.7357$ ROC-AUC) but remains informative, with positive MCC and PR-AUC 0.4520 against a 0.2765 prevalence baseline. Overwriting only `ip_proto` and `tp_src` on those very rows sends ROC-AUC, PR-AUC, balanced accuracy and MCC below their respective no-skill baselines (ROC-AUC 0.3545, MCC −0.1915), while F1 falls to 0.3558. Because CF shares its rows, order, labels, prevalence and seven features with TL, this cannot be attributed to attack mix, labeling, collection environment or sampling; the only variable is the values those two columns hold.

The real ML release lands close to the counterfactual (0.3441 against 0.3545), the ecological half of the argument: because that release also differs in row set and prevalence, it corroborates the direction and magnitude isolated on matched rows rather than re-identifying the effect. The centralized model shows the same pattern ($0.7490 \rightarrow 0.3544 \rightarrow 0.3447$), so this is a property of feature semantics rather than of federated averaging: the hazard is not specific to FedAvg and arises in centralized cross-release evaluation as well.

C. Localizing the reversal

Masking each feature of the fixed model in turn localizes the ranking reversal to protocol: it is the only field whose masking effect changes sign, and it does so on matched rows: on TL it carries the largest transferable contribution, with masking costing 0.1438 ROC-AUC, whereas on CF masking the surrogate *gains* 0.2647. The real release agrees (+0.2103). The KS distance for protocol is 0.3532 on TL and only 0.0992 on CF, so the harmful encoding is the one that looks better aligned, even though the two are computed on the very same rows. Source port shows the converse failure: masking it costs 0.0165 ROC-AUC on TL and its CF distance is maximal because every value is the sentinel, yet masking it leaves ROC-AUC unchanged to four decimals, since a constant column adds the same amount to every logit and cannot reorder scores. It does move the operating point, shifting CF MCC by +0.0186, so protocol localizes the *ranking* reversal rather than the whole fixed-threshold loss.

Retraining confirms the mechanism rather than inferring it from perturbation alone, and yields the comparison that matters when a column is unavailable. Table IV removes fields before training. Dropping protocol costs source F1 ($0.9580 \rightarrow 0.9118$) and collapses the large TL–CF gap, bringing all three views near chance (0.4953, 0.5157, 0.4668), which is consistent with protocol being the dominant driver. The residual differences remain because source-port provenance still differs and ML additionally uses a different row set. Crucially, on the matched rows of CF, *dropping* the unavailable field reaches 0.5157 while *reconstructing* it reaches only 0.3545: incorrect reconstruction is worse than admitting the field is missing, which inverts the intuition behind name-based harmonization. One-hot encoding protocol is the best TL configuration (0.7472) yet barely moves CF (0.3630), so a representation change cannot repair reconstructed values. Removing both ports lowers TL ROC-AUC to 0.7062 with source F1 unchanged; the masking audit attributes that loss primarily to destination port (−0.0605 on TL), which every view supplies, whereas source port is secondary for ranking. Finally, the schema intersection, dropping both unavailable fields rather than reconstructing either, is what a practitioner should prefer: it costs source F1 ($0.9580 \rightarrow 0.9082$) but returns chance-level transfer (0.5075) instead of the 0.3545 reconstruction produces. It also checks

TABLE IV

RETRAINING ABLATIONS, FIVE SEEDS. SOURCE COLUMNS ARE INSDN; VIEW COLUMNS ARE ZERO-SHOT ROC-AUC. THE INTERSECTION ROW KEEPS ONLY THE SEVEN FIELDS BOTH RELEASES SUPPLY AS READ VALUES.

Features	Src. F1	Src. AUC	TL	CF	ML
All nine	0.9580±0.0023	0.9324±0.0039	0.7357	0.3545	0.3441
Without ports	0.9583±0.0030	0.9267±0.0065	0.7062	0.3005	0.2857
Without protocol	0.9118±0.0042	0.8915±0.0067	0.4953	0.5157	0.4668
One-hot protocol	0.9582±0.0023	0.9311±0.0036	0.7472	0.3630	0.3506
Intersection	0.9082±0.0045	0.8954±0.0076	0.5075	0.5075	0.4505

the counterfactual independently, since with both columns removed CF and TL are the same data and agree to every digit (0.5075 ± 0.0198), confirming they differ in those two columns alone.

D. What a label-free preflight check can and cannot catch

The check of Section III-B sees no target label. Among the 27 evaluated column–view pairs it flags two on all five seeds and no others: the source port of CF and of ML, which retain none of their variation (ratio 0.0000) and lie entirely outside the source-training range (out-of-range 1.0000). The separation is not marginal, since no unflagged column falls below ratio 0.352 or exceeds out-of-range 0.0028. The one-sided thresholds matter: the byte-rate column is 31 to 48 times more dispersed yet has ΔAUC 0.0000. Per-column values are in `provenance_screen.csv`. We claim no more than a cheap preflight check: detecting a constant placeholder is easy; what matters is the field it does *not* flag.

That field is the one that reverses the verdict. The reconstructed protocol keeps a plausible dispersion ratio (0.5672 on CF) and never leaves the source range, so both conditions stay silent. This is not because the corruption leaves no statistical trace; it does, and the table shows it. The trace simply fails to distinguish semantic corruption from ordinary shift under the marginal rules we tested. Total-variation distance repeats the KS failure, scoring the harmful encoding as the *closer* one (0.1489 against 0.3532) despite identical rows, and the mass-collapse rule fires on all three views, including uncorrupted TL, where protocol 0 falls from 35% of source mass to 0.04%. A rule firing on the intact view as readily as on the damaged ones cannot discriminate, so we report that negative result rather than tune thresholds.

Placeholder substitution is statistically obvious and a preflight check should reject it, whereas a semantically reconstructed field may retain plausible marginal statistics and evade such checks, so it cannot be reliably inferred from the marginal checks tested here and should be established from provenance metadata recorded at preparation time. Our script therefore records, per input file, whether each field was read or reconstructed.

VI. DISCUSSION

A. What the controls establish

The controls share the feature set, records, optimizer family, class weight and threshold, so their close source-domain scores establish only that this linear baseline suits the sampled InSDN task, not that federation improves accuracy. The non-IID experiment makes a narrower point: under the skew of Section IV-B, mean F1 falls only slightly while balanced accuracy and MCC fall visibly, so reporting all three stops high prevalence from hiding an unfavorable tradeoff. Severe non-IID settings, drift-corrected aggregation [7]–[9], client dropout, privacy accounting and communication cost are outside this experiment.

B. Why provenance-sensitive values drive the matched contrast

The TL–CF contrast identifies the causal effect of this two-field intervention on these rows and these models: the views share rows, order, labels, prevalence and seven of nine features, and are scored by the same models under the same source-fitted scaler and threshold, so the difference in Table III cannot be attributed to attack mix, labeling, collection environment or sampling.

The effect is a reversal rather than a loss because the model applies a coefficient to a field whose target values now encode something else. Protocol becomes a TCP-flag indicator omitting UDP entirely, disagreeing with the original on 30.32% of matched rows, so its contribution is not merely uninformative but anti-aligned with target ranking; the fixed-threshold MCC turns negative too. Rosay et al. independently report wrong protocol detection in these releases [20]; we quantify what such a field does to a verdict. The ablations add the corollary: dropping an unavailable field degrades transfer to near-chance, but reconstructing it incorrectly is considerably worse.

Section V-D states the limit of automation. A preflight check catches placeholder substitution and should be run, but it did not catch the reconstruction here, and none of the marginal rules we tested separates the two protocol encodings even on identical rows. A cross-dataset NIDS result must therefore state which release supplied each feature and which fields were reconstructed rather than read. Absent that record, a below-random result cannot be distinguished from an artifact of harmonization, and neither can a favorable one.

The residual TL degradation admits several causes, among them attack mix, label reduction and the limited expressiveness of nine aggregate features; we do not attribute it to any one.

VII. THREATS TO VALIDITY AND REPRODUCIBILITY

The experiment evaluates tabular flows, not packet traces, and is not an SDN deployment. Five repeated holdouts share a finite source corpus and are not independent sites. The contrast covers one source corpus, one target corpus, one classifier family under two training regimes, and three capture files; whether the reversal generalizes to other release pairs or model families is open. The reconstruction is one plausible rule; another would give other magnitudes. The counterfactual establishes what these two fields do to this model on these rows, not that provenance is the largest source of cross-dataset error. The preflight thresholds were fixed at 0.01, not tuned, and the check is one-directional: a flag is strong evidence of substitution, silence is not evidence of sound provenance. Masking is a label-aware model-reliance diagnostic, not a causal attribution, since correlated features share credit. Binary reduction limits external validity.

VIII. Artifact Availability

The preparation script converts named public source files, records which branch supplied each reconstructed field, publishes SHA-256 checksums for the InSDN file and each target view, and aborts if the counterfactual is not row-matched. The evaluation script regenerates all five-seed outputs under the pinned Python 3.11 environment, and a companion script re-reads this paper and fails if any reported value disagrees with them. Under current ACM terms, the pinned environment, fixed seeds and verifier give repeatability in our setup; the released scripts and checksummed inputs are intended to *enable* independent reproduction on these artifacts, which we have not externally validated; replication from independent artifacts and other release pairs remain open. The artifact is at https://github.com/thtcsec/rivf2026-federated-edge; third-party datasets are not redistributed.

IX. Conclusion

Matched controls show no FedAvg accuracy advantage on sampled InSDN, while non-IID skew primarily harms fixed-threshold measures. On one fixed set of 207,463 target flows with identical labels, prevalence and seven identical features, overwriting only protocol and source port with one `MachineLearningCVE`-compatible harmonization's surrogates moves ROC-AUC from 0.7357 to 0.3545 and MCC from $+0.2511$ to -0.1915. The real release shows the same direction and magnitude (0.3441), as does a centralized model, so this is a property of feature semantics rather than of federated averaging. Masking localizes the ranking reversal to protocol, whose smaller marginal shift is the more harmful encoding, and dropping an unavailable field is safer than reconstructing it (0.5157 versus 0.3545), so the seven-field intersection is our deployment-compatible configuration. A label-free preflight check catches the substituted port but not the reconstructed protocol. Cross-dataset NIDS evaluations should report feature provenance per field. Future work should test other corpora, release pairs, reconstruction rules, and model families.

Acknowledgment

The authors thank the Faculty of Information Technology, Ho Chi Minh City University of Foreign Languages – Information Technology, for academic support. **Generative AI Disclosure**: OpenAI Codex and Cursor were used to draft and revise prose throughout this manuscript and to refactor the evaluation scripts. They generated no experimental data, numerical result, reference, or scientific conclusion: every reported value is produced by the released scripts and re-checked by the accompanying verifier, and every citation was checked against the publisher record. The authors designed the study, ran all code, and reviewed and take responsibility for the final text, code and claims.

References

[1] H. B. McMahan *et al.*, “Communication-Efficient Learning of Deep Networks from Decentralized Data,” in *Proc. AISTATS*, 2017, pp. 1273–1282.

[2] V. Mothukuri *et al.*, “A Survey on Security and Privacy of Federated Learning,” *Future Gener. Comput. Syst.*, vol. 115, pp. 619–640, 2021.

[3] L. D’hooge *et al.*, “Inter-Dataset Generalization Strength of Supervised Machine Learning Methods for Intrusion Detection,” *J. Inf. Secur. Appl.*, vol. 54, Art. 102564, 2020, doi: 10.1016/j.jisa.2020.102564.

[4] G. Apruzzese, L. Pajola, and M. Conti, “The Cross-Evaluation of Machine Learning-Based Network Intrusion Detection Systems,” *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 4, pp. 5152–5169, 2022, doi: 10.1109/TNSM.2022.3157344.

[5] S. Layeghy and M. Portmann, “Explainable Cross-Domain Evaluation of ML-Based Network Intrusion Detection Systems,” *Comput. Electr. Eng.*, vol. 108, Art. 108692, 2023, doi: 10.1016/j.compeleceng.2023.108692.

[6] M. Cantone, C. Marrocco, and A. Bria, “Machine Learning in Network Intrusion Detection: A Cross-Dataset Generalization Study,” *IEEE Access*, vol. 12, pp. 144489–144508, 2024, doi: 10.1109/ACCESS.2024.3472907.

[7] T. Li *et al.*, “Federated Optimization in Heterogeneous Networks,” in *Proc. MLSys*, vol. 2, 2020, pp. 429–450.

[8] S. P. Karimireddy *et al.*, “SCAFFOLD: Stochastic Controlled Averaging for Federated Learning,” in *Proc. ICML*, PMLR vol. 119, 2020, pp. 5132–5143.

[9] J. Wang *et al.*, “Tackling the Objective Inconsistency Problem in Heterogeneous Federated Optimization,” in *Proc. NeurIPS*, 2020.

[10] G. de Carvalho Bertoli, L. A. Pereira Júnior, and O. Saotome, “Improving Detection of Scanning Attacks on Heterogeneous Networks with Federated Learning,” *ACM SIGMETRICS Perform. Eval. Rev.*, vol. 49, no. 4, pp. 118–123, 2022, doi: 10.1145/3543146.3543172.

[11] T. E. T. Djaidja *et al.*, “Federated Learning for 5G and Beyond, a Blessing and a Curse–An Experimental Study on Intrusion Detection Systems,” *Comput. Secur.*, vol. 139, Art. 103707, 2024, doi: 10.1016/j.cose.2024.103707.

[12] A. Belenguer, J. A. Pascual, and J. Navaridas, “A Review of Federated Learning Applications in Intrusion Detection Systems,” *Comput. Netw.*, vol. 258, Art. 111023, 2025, doi: 10.1016/j.comnet.2024.111023.

[13] N. H. Quyen *et al.*, “FedKD-IDS: A Robust Intrusion Detection System Using Knowledge Distillation-Based Semi-Supervised Federated Learning and Anti-Poisoning Attack Mechanism,” *Inf. Fusion*, vol. 117, Art. 102807, 2025, doi: 10.1016/j.inffus.2024.102807.

[14] M. A. Bilal *et al.*, “Dataset-Centric Evaluation of Federated Intrusion Detection Models in IoT Networks,” *Sci. Rep.*, vol. 16, Art. 2683, 2026, doi: 10.1038/s41598-025-32567-w.

[15] M. Ring *et al.*, “A Survey of Network-Based Intrusion Detection Data Sets,” *Comput. Secur.*, vol. 86, pp. 147–167, 2019, doi: 10.1016/j.cose.2019.06.005.

[16] A. Demirbaş Paray and M. Aydos, “Federated Learning for Intrusion Detection Under Class Imbalance: A Multi-Domain Ablation Study with Per-Client SMOTE,” *Appl. Sci.*, vol. 16, no. 2, Art. 801, 2026, doi: 10.3390/app16020801.

[17] M. Sarhan, S. Layeghy, and M. Portmann, “Evaluating Standard Feature Sets Towards Increased Generalisability and Explainability of ML-Based Network Intrusion Detection,” *Big Data Res.*, vol. 30, Art. 100359, 2022, doi: 10.1016/j.bdr.2022.100359.

[18] G. Engelen, V. Rimmer, and W. Joosen, “Troubleshooting an Intrusion Detection Dataset: The CICIDS2017 Case Study,” in *Proc. IEEE SPW*, 2021, pp. 7–12, doi: 10.1109/SPW53761.2021.00009.

[19] M. Lanvin *et al.*, “Errors in the CICIDS2017 Dataset and the Significant Differences in Detection Performances It Makes,” in *Proc. CRiSIS 2022*, Springer, 2023, pp. 18–33, doi: 10.1007/978-3-031-31108-6_2.

[20] A. Rosay *et al.*, “Network Intrusion Detection: A Comprehensive Analysis of CIC-IDS2017,” in *Proc. ICISSP*, 2022, pp. 25–36, doi: 10.5220/0010774000003120.

[21] M. Chizari, Q. Khan Ali Mirza, A. Alam, and H. Chizari, “Closed-Set Heterogeneous Domain Adaptation for IoT Intrusion Detection: An Anchor-Based Benchmark Across Single- and Multi-Source Transfer,” *Sensors*, vol. 26, no. 11, Art. 3610, 2026, doi: 10.3390/s26113610.

[22] M. S. Elsayed, N.-A. Le-Khac, and A. D. Jurcut, “InSDN: A Novel SDN Intrusion Dataset,” *IEEE Access*, vol. 8, pp. 165263–165284, 2020, doi: 10.1109/ACCESS.2020.3022633.

[23] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, “Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization,” in *Proc. ICISSP*, 2018, pp. 108–116, doi: 10.5220/0006639801080116.